Case Study 7: Inventory and Warehouse Management System

Part 1: Requirement Analysis and Database Design

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1. Problem overview

A multinational manufacturing company tracks its inventory in spreadsheets. It wants a database system that manages products, categories, suppliers, warehouses, inventory, purchase orders, purchase items, goods receipts, stock transfers, dispatches and stock adjustments. The system must show live stock levels and support procurement analysis, warehouse utilization, stock movement tracking and management reports.

The design below keeps the number of tables small (13 relations). Every table maps to one element of the ER diagram, so the diagram and the database match one to one.

2. Functional requirements

ID	Requirement	Description
FR1	Manage products and categories	Add, update, delete and search products. Each product is placed in one category.
FR2	Manage suppliers	Store supplier details, a contact person, email and one or more phone numbers.
FR3	Manage warehouses	Store each warehouse with its location (city, country) and storage capacity.
FR4	Track inventory	Keep the current quantity of every product in every warehouse.
FR5	Create purchase orders	Raise a purchase order to one supplier with one or more purchase items, and track its status.
FR6	Record goods receipts	Record goods received against a purchase item. Allow partial deliveries and record rejected quantity. Stock goes up.
FR7	Record stock transfers	Move stock from one warehouse to another. Stock goes down at the source and up at the destination.
FR8	Record dispatches	Record goods sent out to customers. Stock goes down.
FR9	Record stock adjustments	Correct stock for damage, loss, expiry or a physical count difference, with a reason.
FR10	Real-time inventory monitoring	Show stock on hand per product and warehouse, and flag products at or below the reorder level.
FR11	Procurement analysis	Report spending per supplier, pending and partially received orders, and ordered vs received quantity.
FR12	Warehouse utilization	Compare total stock in each warehouse with its capacity.

ID	Requirement	Description
FR13	Stock movement tracking	Show the full history of stock changes for any product or warehouse.
FR14	Management reporting	Produce summary reports (stock value, top moving products, adjustment losses) from the stored data.

3. Business rules

ID	Business rule
BR1	Every product belongs to exactly one category. A category can have zero or many products.
BR2	Each product has a unique SKU (stock keeping unit code).
BR3	There is only one inventory record for each product in each warehouse. Quantity on hand can never be negative.
BR4	When quantity on hand is at or below the product's reorder level, the product is shown as low stock.
BR5	Total stock stored in a warehouse must not go above the warehouse capacity.
BR6	A purchase order is placed with exactly one supplier. A supplier can receive many purchase orders.
BR7	A purchase order must contain at least one purchase item. Line numbers are unique only inside their own order.
BR8	Ordered quantity must be greater than zero. Unit cost cannot be negative.
BR9	The order total is not stored. It is calculated as the sum of Qty_Ordered x Unit_Cost of its items.
BR10	Goods may arrive in parts, so one purchase item can have many goods receipts. The total received cannot be more than the ordered quantity, and the received product must be the same as the ordered product.
BR11	Rejected quantity is not added to stock. Only the accepted quantity increases inventory.
BR12	Every change to stock is saved as a stock movement. Each movement is exactly one type: goods receipt, dispatch, stock adjustment or stock transfer.
BR13	A dispatch or a transfer cannot take out more stock than is available in the source warehouse.
BR14	A stock transfer must go to a different warehouse than the source warehouse.
BR15	Every adjustment needs a reason (Damage, Loss, Expiry, Count correction) and a direction (Increase or Decrease).
BR16	Purchase order status is one of: Pending, Approved, Partially Received, Received, Cancelled.

Effect of each movement type on stock

Movement type	Source warehouse (Warehouse_ID)	Destination warehouse
Goods receipt	+ Quantity	-
Dispatch	- Quantity	-
Stock adjustment	+ Quantity (Increase) or - Quantity (Decrease)	-
Stock transfer	- Quantity	+ Quantity at To_Warehouse_ID

4. Entities and attributes

4.1 Case study items and where they appear in the design

Case study item	Design element	Kind
Products	PRODUCT	Strong entity
Categories	CATEGORY	Strong entity
Suppliers	SUPPLIER	Strong entity
Warehouses	WAREHOUSE	Strong entity
Inventory	STOCKED_IN (aggregated as [INVENTORY])	M:N relationship between PRODUCT and WAREHOUSE
Purchase orders	PURCHASE_ORDER	Strong entity
Purchase items	PURCHASE_ITEM	Weak entity, owner is PURCHASE_ORDER
Goods receipts	GOODS_RECEIPT	Subclass of STOCK_MOVEMENT
Stock transfers	STOCK_TRANSFER	Subclass of STOCK_MOVEMENT
Dispatches	DISPATCH	Subclass of STOCK_MOVEMENT
Stock adjustments	STOCK_ADJUSTMENT	Subclass of STOCK_MOVEMENT
(added for tracking)	STOCK_MOVEMENT	Superclass. Holds the common data of all four movement types

4.2 Entity types and their attributes

Underlined = key attribute. Dashed underline in the diagram = partial key. { } = multivalued. Attributes in ( ) under a name are parts of a composite attribute.

Entity	Type	Attributes	Identifier
CATEGORY	Strong	Category_ID, Category_Name, Description	Category_ID
PRODUCT	Strong	Product_ID, Product_Name, SKU, Unit_Price, Reorder_Level	Product_ID (SKU is also unique)
SUPPLIER	Strong	Supplier_ID, Supplier_Name, Contact_Person, Email, {Phone_No} (multivalued)	Supplier_ID
WAREHOUSE	Strong	Warehouse_ID, Warehouse_Name, Capacity, Location (City, Country) (composite)	Warehouse_ID
PURCHASE_ORDER	Strong	PO_ID, Order_Date, Expected_Date, Status, Total_Amount (derived)	PO_ID
PURCHASE_ITEM	Weak (owner: PURCHASE_ORDER)	Line_No (partial key), Qty_Ordered, Unit_Cost	PO_ID + Line_No
STOCK_MOVEMENT	Superclass	Movement_ID, Movement_Date, Quantity, Movement_Type (discriminator)	Movement_ID
GOODS_RECEIPT	Subclass	Qty_Rejected (plus all STOCK_MOVEMENT attributes)	Movement_ID (inherited)
DISPATCH	Subclass	Customer_Name, Destination	Movement_ID (inherited)
STOCK_ADJUSTMENT	Subclass	Reason, Adjust_Type	Movement_ID (inherited)

Entity	Type	Attributes	Identifier
STOCK_TRANSFER	Subclass	Transfer_Status	Movement_ID (inherited)
STOCKED_IN [INVENTORY]	Relationship (aggregated)	Qty_On_Hand, Last_Updated	Product_ID + Warehouse_ID

4.3 Design decisions

Decision	Reason
Inventory as a relationship	Inventory has no identity of its own. It is the quantity of one product in one warehouse, so it is the M:N relationship STOCKED_IN with the attributes Qty_On_Hand and Last_Updated.
Aggregation [INVENTORY]	A stock movement changes the stock of one product in one warehouse, which is one STOCKED_IN instance. To relate STOCK_MOVEMENT to that pair, STOCKED_IN is treated as a higher-level entity (aggregation).
Purchase item as a weak entity	A purchase item cannot exist without its order. It is identified by the order's PO_ID plus its own Line_No.
Movements as a specialization	Receipts, dispatches, adjustments and transfers all have a date, a quantity, a product and a warehouse. These common attributes sit in STOCK_MOVEMENT, and each subclass keeps only its own extra data. The specialization is disjoint (a movement has one type) and total (every movement has a type). One table now holds the full stock history.
Derived total	Total_Amount of a purchase order is derived from its items, so it is not stored. This avoids update errors.

5. Relationships and cardinalities

The (min,max) pair is written next to the entity it belongs to. It tells how many times one instance of that entity takes part in the relationship. min = 1 means total participation (double line in the diagram).

Relationship	Entities with (min,max)	Ratio	Participation	Meaning
BELONGS_TO	CATEGORY (0,N) : PRODUCT (1,1)	1 : N	Product side total	A category groups many products. A product is in one category.
STOCKED_IN	PRODUCT (0,N) : WAREHOUSE (0,N)	M : N	Both partial	A product can be stored in many warehouses and a warehouse stores many products. Attributes: Qty_On_Hand, Last_Updated.
PLACED_WITH	SUPPLIER (0,N) : PURCHASE_ORDER (1,1)	1 : N	Order side total	Each order goes to exactly one supplier.
CONTAINS (identifying)	PURCHASE_ORDER (1,N) : PURCHASE_ITEM (1,1)	1 : N	Both total	An order has at least one item. An item belongs to one order.
ORDERS	PRODUCT (0,N) : PURCHASE_ITEM (1,1)	1 : N	Item side total	Each purchase item orders one product.
RECEIVED_AGAINST	PURCHASE_ITEM (0,N) : GOODS_RECEIPT (1,1)	1 : N	Receipt side total	One item can be received in several parts. Each receipt is for one item.
AFFECTS	[INVENTORY] (0,N) : STOCK_MOVEMENT (1,1)	1 : N	Movement side total	Each movement changes one product's stock in one warehouse.

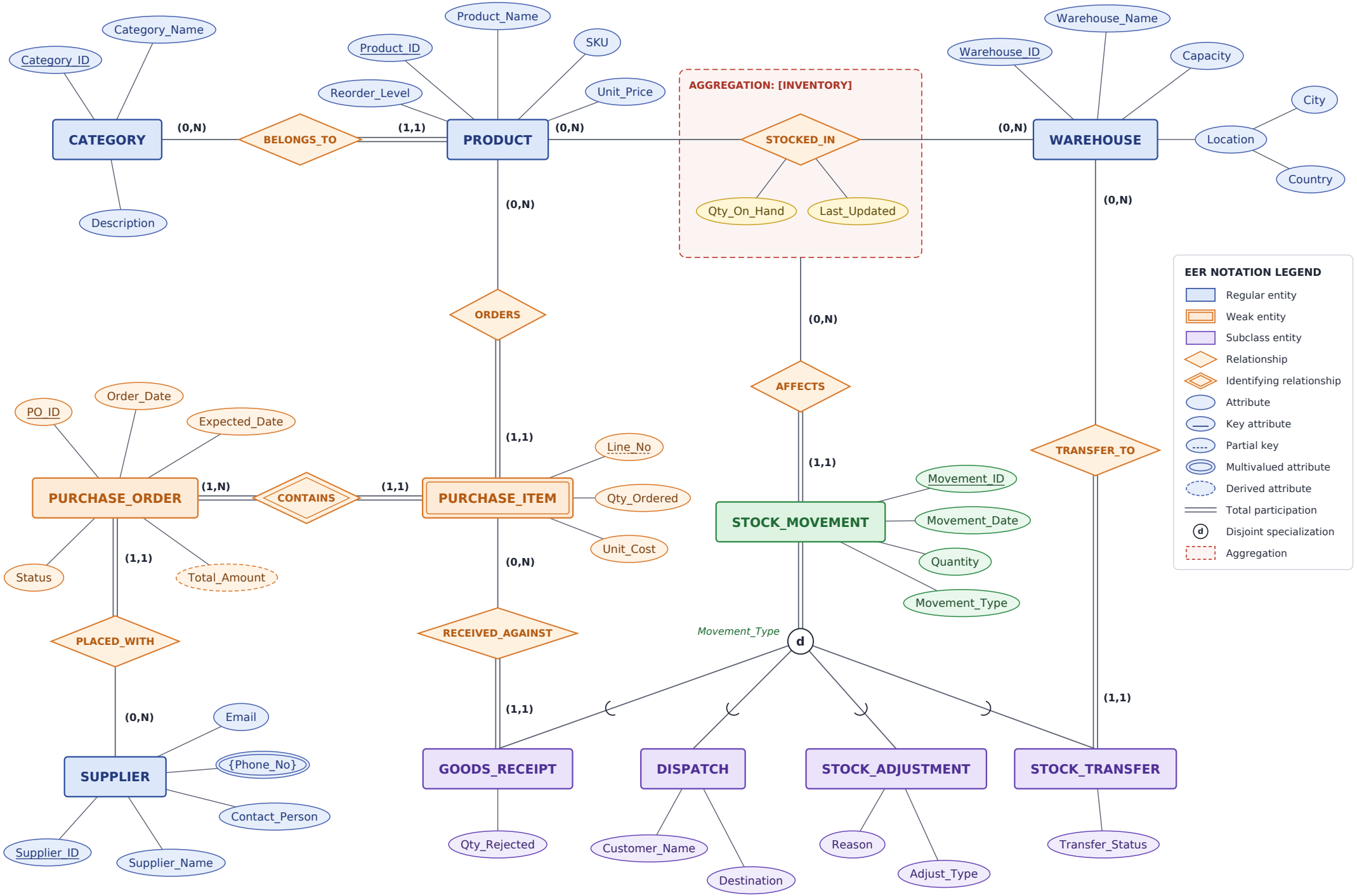
Relationship	Entities with (min,max)	Ratio	Participation	Meaning
TRANSFER_TO	WAREHOUSE (0,N) : STOCK_TRANSFER (1,1)	1 : N	Transfer side total	Each transfer has one destination warehouse.
Specialization (d)	STOCK_MOVEMENT to GOODS_RECEIPT, DISPATCH, STOCK_ADJUSTMENT, STOCK_TRANSFER	IS-A	Disjoint, total	Discriminator attribute: Movement_Type.

6. ER diagram

The EER diagram is on the next page (landscape). It uses the same notation as the reference diagram: rectangles for entities, double rectangles for weak entities, diamonds for relationships, double diamonds for identifying relationships, ovals for attributes, a circle with d for disjoint specialization and a dashed box for aggregation. Every entity, relationship and attribute in the diagram appears in the relational schema in Section 8.

6. ER Diagram: Inventory and Warehouse Management System (EER)

Weak entity, composite / multivalued / derived attributes, aggregation [INVENTORY], disjoint total specialization and (min,max) constraints



7. ER to relational mapping

EER element	Mapping rule	Result in this design
Strong entities	One table each. The key attribute becomes the primary key.	CATEGORY, PRODUCT, SUPPLIER, WAREHOUSE, PURCHASE_ORDER
Composite attribute	Keep only the simple parts as columns.	Location becomes City, Country in WAREHOUSE
Multivalued attribute	New table with the owner key plus the value. Both together form the primary key.	SUPPLIER_PHONE(Supplier_ID, Phone_No)
Derived attribute	Not stored. Calculated in a query when needed.	Total_Amount = SUM(Qty_Ordered x Unit_Cost)
Weak entity	New table. Primary key = owner key + partial key. Owner key is also a foreign key.	PURCHASE_ITEM(PO_ID, Line_No, ...)
1 : N relationship	Put the primary key of the 1 side as a foreign key in the N side table.	Category_ID in PRODUCT, Supplier_ID in PURCHASE_ORDER, Product_ID in PURCHASE_ITEM, (PO_ID, Line_No) in GOODS_RECEIPT, To_Warehouse_ID in STOCK_TRANSFER
M : N relationship	New table with both primary keys as a composite primary key, plus the relationship attributes.	INVENTORY(Product_ID, Warehouse_ID, Qty_On_Hand, Last_Updated)
Aggregation	The aggregated relationship's table acts as an entity. The 1 : N AFFECTS relationship adds its key as a foreign key.	(Product_ID, Warehouse_ID) in STOCK_MOVEMENT references INVENTORY
Specialization (disjoint, total)	One table for the superclass and one per subclass. Subclass primary key = superclass key, also a foreign key.	STOCK_MOVEMENT, GOODS_RECEIPT, DISPATCH, STOCK_ADJUSTMENT, STOCK_TRANSFER

8. Relational schema

8.1 Simplified relations

Underlined = primary key. Italic = foreign key. A column can be both.

1. CATEGORY (Category_ID, Category_Name, Description)
2. PRODUCT (Product_ID, Product_Name, SKU, Unit_Price, Reorder_Level, *Category_ID*)
3. WAREHOUSE (Warehouse_ID, Warehouse_Name, City, Country, Capacity)
4. INVENTORY (*Product_ID*, *Warehouse_ID*, Qty_On_Hand, Last_Updated)
5. SUPPLIER (Supplier_ID, Supplier_Name, Contact_Person, Email)
6. SUPPLIER_PHONE (*Supplier_ID*, Phone_No)
7. PURCHASE_ORDER (PO_ID, Order_Date, Expected_Date, Status, *Supplier_ID*)
8. PURCHASE_ITEM (*PO_ID*, Line_No, *Product_ID*, Qty_Ordered, Unit_Cost)
9. STOCK_MOVEMENT (Movement_ID, Movement_Date, Quantity, Movement_Type, *Product_ID*, *Warehouse_ID*)
10. GOODS_RECEIPT (*Movement_ID*, *PO_ID*, *Line_No*, Qty_Rejected)
11. DISPATCH (*Movement_ID*, Customer_Name, Destination)
12. STOCK_ADJUSTMENT (*Movement_ID*, Reason, Adjust_Type)
13. STOCK_TRANSFER (*Movement_ID*, *To_Warehouse_ID*, Transfer_Status)

8.2 Table structures

Table 1: CATEGORY

Column	Data type	Constraints
Category_ID	INT	PRIMARY KEY
Category_Name	VARCHAR(50)	NOT NULL, UNIQUE
Description	VARCHAR(200)	

Table 2: PRODUCT

Column	Data type	Constraints
Product_ID	INT	PRIMARY KEY
Product_Name	VARCHAR(100)	NOT NULL
SKU	VARCHAR(30)	NOT NULL, UNIQUE
Unit_Price	DECIMAL(10,2)	CHECK (Unit_Price >= 0)
Reorder_Level	INT	DEFAULT 0, CHECK (Reorder_Level >= 0)
Category_ID	INT	NOT NULL, FOREIGN KEY references CATEGORY(Category_ID)

Table 3: WAREHOUSE

Column	Data type	Constraints
Warehouse_ID	INT	PRIMARY KEY
Warehouse_Name	VARCHAR(100)	NOT NULL
City	VARCHAR(50)	NOT NULL
Country	VARCHAR(50)	NOT NULL
Capacity	INT	NOT NULL, CHECK (Capacity > 0)

Table 4: INVENTORY

Column	Data type	Constraints
Product_ID	INT	PRIMARY KEY (part), FOREIGN KEY references PRODUCT(Product_ID)
Warehouse_ID	INT	PRIMARY KEY (part), FOREIGN KEY references WAREHOUSE(Warehouse_ID)
Qty_On_Hand	INT	NOT NULL, DEFAULT 0, CHECK (Qty_On_Hand >= 0)
Last_Updated	DATETIME	NOT NULL

Table 5: SUPPLIER

Column	Data type	Constraints
Supplier_ID	INT	PRIMARY KEY
Supplier_Name	VARCHAR(100)	NOT NULL
Contact_Person	VARCHAR(100)	
Email	VARCHAR(100)	UNIQUE

Table 6: SUPPLIER_PHONE

Column	Data type	Constraints
Supplier_ID	INT	PRIMARY KEY (part), FOREIGN KEY references SUPPLIER(Supplier_ID)
Phone_No	VARCHAR(15)	PRIMARY KEY (part)

Table 7: PURCHASE_ORDER

Column	Data type	Constraints
PO_ID	INT	PRIMARY KEY
Order_Date	DATE	NOT NULL
Expected_Date	DATE	CHECK (Expected_Date >= Order_Date)
Status	VARCHAR(20)	NOT NULL, CHECK (Status IN ('Pending', 'Approved', 'Partially Received', 'Received', 'Cancelled'))
Supplier_ID	INT	NOT NULL, FOREIGN KEY references SUPPLIER(Supplier_ID)

Table 8: PURCHASE_ITEM

Column	Data type	Constraints
PO_ID	INT	PRIMARY KEY (part), FOREIGN KEY references PURCHASE_ORDER(PO_ID) ON DELETE CASCADE
Line_No	INT	PRIMARY KEY (part)
Product_ID	INT	NOT NULL, FOREIGN KEY references PRODUCT(Product_ID)
Qty_Ordered	INT	NOT NULL, CHECK (Qty_Ordered > 0)
Unit_Cost	DECIMAL(10,2)	NOT NULL, CHECK (Unit_Cost >= 0)

Table 9: STOCK_MOVEMENT

Column	Data type	Constraints
Movement_ID	INT	PRIMARY KEY
Movement_Date	DATETIME	NOT NULL
Quantity	INT	NOT NULL, CHECK (Quantity > 0)
Movement_Type	VARCHAR(12)	NOT NULL, CHECK (Movement_Type IN ('RECEIPT', 'DISPATCH', 'ADJUSTMENT', 'TRANSFER'))
Product_ID	INT	NOT NULL
Warehouse_ID	INT	NOT NULL. (Product_ID, Warehouse_ID) is a FOREIGN KEY references INVENTORY(Product_ID, Warehouse_ID)

Table 10: GOODS_RECEIPT

Column	Data type	Constraints
Movement_ID	INT	PRIMARY KEY, FOREIGN KEY references STOCK_MOVEMENT(Movement_ID)
PO_ID	INT	NOT NULL
Line_No	INT	NOT NULL. (PO_ID, Line_No) is a FOREIGN KEY references PURCHASE_ITEM(PO_ID, Line_No)
Qty_Rejected	INT	DEFAULT 0, CHECK (Qty_Rejected >= 0)

Table 11: DISPATCH

Column	Data type	Constraints
Movement_ID	INT	PRIMARY KEY, FOREIGN KEY references STOCK_MOVEMENT(Movement_ID)
Customer_Name	VARCHAR(100)	NOT NULL
Destination	VARCHAR(150)	NOT NULL

Table 12: STOCK_ADJUSTMENT

Column	Data type	Constraints
Movement_ID	INT	PRIMARY KEY, FOREIGN KEY references STOCK_MOVEMENT(Movement_ID)
Reason	VARCHAR(20)	NOT NULL, CHECK (Reason IN ('Damage', 'Loss', 'Expiry', 'Count correction'))
Adjust_Type	VARCHAR(10)	NOT NULL, CHECK (Adjust_Type IN ('Increase', 'Decrease'))

Table 13: STOCK_TRANSFER

Column	Data type	Constraints
Movement_ID	INT	PRIMARY KEY, FOREIGN KEY references STOCK_MOVEMENT(Movement_ID)
To_Warehouse_ID	INT	NOT NULL, FOREIGN KEY references WAREHOUSE(Warehouse_ID)
Transfer_Status	VARCHAR(15)	NOT NULL, CHECK (Transfer_Status IN ('In transit', 'Completed'))

9. Primary and foreign keys

Table	Primary key	Foreign key(s) → referenced table
CATEGORY	Category_ID	-
PRODUCT	Product_ID	Category_ID → CATEGORY
WAREHOUSE	Warehouse_ID	-
INVENTORY	(Product_ID, Warehouse_ID)	Product_ID → PRODUCT Warehouse_ID → WAREHOUSE
SUPPLIER	Supplier_ID	-
SUPPLIER_PHONE	(Supplier_ID, Phone_No)	Supplier_ID → SUPPLIER
PURCHASE_ORDER	PO_ID	Supplier_ID → SUPPLIER
PURCHASE_ITEM	(PO_ID, Line_No)	PO_ID → PURCHASE_ORDER Product_ID → PRODUCT
STOCK_MOVEMENT	Movement_ID	(Product_ID, Warehouse_ID) → INVENTORY
GOODS_RECEIPT	Movement_ID	Movement_ID → STOCK_MOVEMENT (PO_ID, Line_No) → PURCHASE_ITEM
DISPATCH	Movement_ID	Movement_ID → STOCK_MOVEMENT
STOCK_ADJUSTMENT	Movement_ID	Movement_ID → STOCK_MOVEMENT
STOCK_TRANSFER	Movement_ID	Movement_ID → STOCK_MOVEMENT To_Warehouse_ID → WAREHOUSE

Candidate keys: SKU in PRODUCT, Category_Name in CATEGORY and Email in SUPPLIER are also unique, so each is a candidate key. They are not chosen as primary keys because a short integer ID never changes and keeps the foreign keys small.

10. Requirement coverage check

Each reporting need from the case study is answered by the tables below, so no extra table is needed.

Need	Tables used	How
Real-time inventory monitoring	INVENTORY, PRODUCT, WAREHOUSE	Qty_On_Hand per product and warehouse. Low stock when Qty_On_Hand <= Reorder_Level.
Procurement analysis	PURCHASE_ORDER, PURCHASE_ITEM, GOODS_RECEIPT, STOCK_MOVEMENT, SUPPLIER	Spend per supplier = SUM(Qty_Ordered x Unit_Cost). Ordered vs received = Qty_Ordered compared with SUM(Quantity) of its receipts.
Warehouse utilization	INVENTORY, WAREHOUSE	SUM(Qty_On_Hand) / Capacity for each warehouse.
Stock movement tracking	STOCK_MOVEMENT and its four subclass tables	All stock changes in date order, filtered by product, warehouse or Movement_Type.
Management reporting	All tables	Stock value (Qty_On_Hand x Unit_Price), top dispatched products, losses from adjustments.